Zhengwei Zhang

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Summary

Health Data Science graduate student at UCSF with research experience in biomedical AI, statistical modeling, and translational data science. Experienced in applying machine learning and quantitative methods to clinical, epidemiological, imaging, and physiological data using Python and R. Interested in computational biology and cancer research, with a focus on integrating diverse biomedical data to better understand disease processes and support precision medicine.

Technical Skills

- **Programming Languages**: Python (PyTorch, Scikit-learn, Pandas, Numpy), R, SQL, C/C++
- **Software**: IBM SPSS, MATLAB, MS Office
- **General Skills**: Git, LaTeX, GPU-based model training, Linux Command Line

Education

University of California, San Francisco, USA Jul 2024 - Jun 2026
Master of Science in Health Data Science

- GPA: 3.8/4.0
- **Related Coursework:** Machine Learning, Deep Learning, Applied Data Science with Python, Biostatistical Methods

Hangzhou Dianzi University, Hangzhou, China Sep 2020 - Jun 2024
Bachelor of Engineering in Biomedical Information Engineering

- **Related Coursework:** Calculus-based Physics, Biochemistry, Cell and Molecular Biology, Digital Signal Processing, Digital Image Processing, Analog Circuits

Research Project

eGene Program (Capstone Project) Jun 2025 - Jun 2026
University of California, San Francisco
Advisor: Li Zhang, PhD; Project PI: Pamela Munster, MD

- Analyzed a large-scale clinical and epidemiological dataset comprising 1,603 participants and 216 variables to investigate determinants of cancer risk among individuals carrying hereditary cancer-associated mutations.
- Developed statistical models in R, including logistic regression, LASSO-based feature selection, and gene-environment interaction analyses, to identify risk factors and characterize risk heterogeneity across genetic subgroups.
- Evaluated gene-environment interactions and risk heterogeneity across mutation-defined subgroups to identify factors associated with cancer risk.

Dynamic Multi-Scale Image Fusion for Breast Cancer Diagnosis Apr 2025 - Jun 2025
University of California, San Francisco

- Developed a dynamic multi-scale fusion deep learning model with attention mechanisms for breast cancer classification.
- Developed and evaluated deep learning pipelines for medical image analysis and predictive modeling, achieving an accuracy of 0.87 and an AUC of 0.92.

Vascular Segmentation Program Jun 2022 - Jan 2024
Hangzhou Dianzi University School of Automation
Advisor: Ping Xu, PhD

- Processed and analyzed cardiovascular imaging datasets, including data preprocessing, feature extraction, and visualization for downstream research analysis.
- Developed and evaluated deep learning models for vascular image segmentation using PyTorch, achieving a Dice score of 0.84 and mIoU of 0.76.
- Supported research analysis and presentations through data visualization and figure generation.

Algorithms in Dynamic Blood Glucose Monitoring Oct 2023 - May 2024
Hangzhou Dianzi University School of Automation
Advisor: Kai Fan, PhD

- Developed a personalized continuous glucose monitoring (CGM) framework using Bidirectional LSTM networks and Dynamic Time Warping (DTW) for physiological time-series prediction and alignment.

- Developed anomaly detection algorithms to reduce false alarms and improve monitoring reliability in continuous physiological data.
- Achieved a significant improvement in monitoring accuracy and alarm precision compared to traditional methods, validating the model's clinical and research value.

Publications

Zhang Z, Romanens-Renard L, Zhang L, Munster PN. *Determinants of Cancer Risk in Hereditary Cancer-Prone Individuals: The eGene Study.* Poster Presentation, American Association for Cancer Research Annual Meeting 2026.

Zhang F, Sun H, Xie S, Dong C, Li Y, Xu Y, **Zhang Z**, Chen F. *A tea bud segmentation, detection and picking point localization based on the MDY7-3PTB model.* Frontiers in Plant Science, Sep 2023.

- Co-author; Contributed to result analysis and visualization.

Professional Experience

Teaching Assistant Jan 2026 - Mar 2026
University of California, San Francisco
Course: Machine Learning in R for Biomedical Sciences; Instructor: Adam Olshen, PhD

- Led lab sessions and assisted students with machine learning concepts and R-based data analysis.
- Guided assignments involving statistical modeling and data analysis.
- Supported course instruction by preparing teaching materials and debugging code examples.

Catalyst Program Intern Jun 2025 - Aug 2025
University of California, San Francisco

- Analyzed the economic viability and competitive landscape of a novel Parkinson's treatment.
- Conducted analysis of clinical and regulatory aspects of Parkinson's treatments in a translational research context.

Awards

Scholarship: top 15% AY 2022 - 2023
National-level Model Aircraft Design Competition: Second Prize Oct 2021
Interdisciplinary Contest in Modeling (ICM): Successful Participant Jan 2023